C. 360°

D. cannot determined without actual value of n

35. A man covers some distance in 2 hours. Had he move slower by 4 km/h, he will cover the same distance in 6 hours. Find the speed of man.

A. 4.5 km/h
B. 6 km/h
C. 9 km/h
D. 13.5 km/h

36. The height of a tree was 4.8 m. After two years the height of the tree was increased by 12.5%. The new height of tree after one year is _____ if the height increased with same ratio.

A. 0.6 m
B. 5.1 m
C. 5.4 m
D. 6.0 m

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Four lettered pairs (A to D) follow a rel
words given in the question. Select the
pair that best expresses a relationship
that expressed in the original pair in ca
letters.

16. FLOWER: BUNCH::

A. captain: team
B. man: men
C. manager: business
D. ship: fleet

17. BLACKSMITH: IRON::

A. cook: kitchen
B. farmer: grower
C. potter: clay
D. shoemaker: cobbler

COW: CALF::

A. dog: bark
B. fish: water
C. horse: foal
D. tortoise: shell

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option, from the given ... below each.

10. Emirates airlines plane _____ ten minutes ago.

A. has landed
B. land
C. landed
D. was landed

11. Someone must deal with the situation _____ the spot.

A. beside
B. by
C. of
D. on

12. Though the doors were closed but we could still hear the _____ of laughter.

A. conversation
B. roars
C. voices
D. whispers

13. Which team _____ the game?

A. did it win
B. did they win
C. won
D. won it

14. I'll be back as soon as I _____ my clothes.

A. change
B. changed
C. changes
D. have changed

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42. What is the angle between the hands of a clock at 9 O'clock?

A. 0°
B. 60°
C. 90°
D. 120°

43. In a scalene triangle _____.

A. all angles are different in measure
B. all angles are same in measure
C. every angle is of 90°
D. two angles are same in measure

44. A dice is rolled twice. The probability of getting doublets is _____.

A. 1/36
B. 1/18
C. 1/9
D. 1/6

45. The value of x in the linear equation $\frac{3x-4}{2} - \frac{x}{4} = \frac{13}{6}$ is _____.

A. -50/3
B. -10/3
C. 10/3
D. 50/3

46. Find the quadratic equation whose roots are 1 and 2.

A. $x^2 - 3x + 2$
B. $x^2 - 3x - 2 = 0$
C. $x^2 - 3x + 2 = 0$
D. $x^2 - 2x + 3 = 0$

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16. Which of the following is a perfect number

A. 24
B. 26
C. 28
D. 32

2, 2, 2, 2, 2, 2, 2, 2, 2 i

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40. $3\log_{10}5 + \log_{10}24 - \frac{1}{2}\log_{10}9 =$ _____.

A. 2
B. 3
C. 100
D. 1000

$5\sqrt{5}$

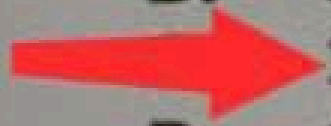
360
130
230

360
180
180
0

10/

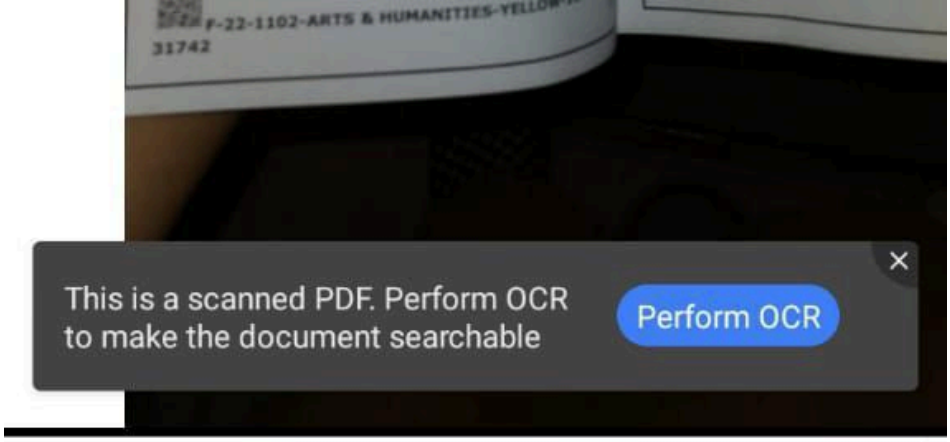
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34. The number of 5-digit numbers, that can be formed from the digits 4, 3, 5, 7, 9 is _____.

A. 60
B. 120

140
D. 180

is 90° are called _____

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25. If the cost of 2 copies and 3 books is Rs. 1300 and cost 5 copies and 1 book is 2600. Find the cost of a copy and a book.

A. Rs. 100
B. Rs. 500
C. Rs. 600
D. Rs. 50000

26. Abdullah gave Rs. 150 to Umar and Umar take Rs. 65 to Ahmed. After this all of them have same money. How much more Abdullah has than Ahmed?

A. Rs. 30
B. Rs. 35
C. Rs. 85
D. Rs. 215

27. Ahmed added to containers of paint containing 40 liters and 55 liters of paint. After using 49 liters in room and 34 liters in kitchen, how much paint is left?

A. 7 liters
B. 12 liters
C. 21 liters
D. 22 liters

28. If f(x) = 5a then it is called _____ function.

A. constant
B. identity
C. one-one
D. quadratic

29. The quadratic equation with roots 2, -3 is _____.

A. $x^2 - x + 6$
B. $x^2 + x + 6 = 0$
C. $x^2 + x - 6 = 0$
D. $x^2 - x - 6 = 0$

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29. A certain number of balls were purchased for Rs. 750. Five more balls will be purchased with the same amount if price of each ball is less by Rs. 12.5. The original number of purchased balls is _____.

21.

- A. 10
- B. 15
- C. 20
- D. 50

$-1 = 9 - 15 - 70$

$-1 = 6 - 70$

$70 \quad -70$
$6 \quad -15$
$64 \quad -85$

30. If $f(x) = x^2 + 5kx - 70$ and $f(-3) = -1$ then value of k is _____.

- A. -62/15
- B. -4
- C. 4
- D. 62/15

$-1 = 9 - 15 - 70$

$= 9 - 85$

-85
9
76

31. Abdullah gave Rs. 135 to Umar, and Umar takes Rs. 65 from Ahmed. After this all of them have same amount of money. How much Abdullah has more than Ahmed?

- A. Rs. 65
- B. Rs. 70
- C. Rs. 265
- D. Rs. 335

$a = s + t$

$d = 10 + 8 + 7$

135
65
70

$s = \frac{d}{t}$

$s = \frac{\quad}{8}$

32. A man moves from town A to town B in 8 hours. Coming back from B to A he moves faster by 10 km/h and reach to town A in 7 hours. What is the distance from town A to B?

- A. 56 km
- B. 420 km
- C. 560 km
- D. 1120 km

$(a+b)^2 = a^2 + 2ab + b^2$

$d = 10 \times 15$

$600 -$

600
16
584

33. If $a + b = 10$ and $a - b = 4$, then the value of $2(a^2 - b^2)$ is _____.

- A. 20
- B. 40
- C. 60
- D. 80

$(a+b)^2 - (a-b)^2 =$

$62 \quad -1+61$

$-3 = 1 - 5R - 70$

69
3
66

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D. 6

44. If A:B = 5:7 and C:B = 7:11 then A:B:C = _____

A. 49:45:77
B. 35:55:66
C. 55:77:49
D. 49:55:77

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59. If the solubility of PbF_2 at 25°C is 0.64g/dm^3, what will be its K_{sp} value?

A. $4.0x10^{-8}$
B. $6.0x10^{-8}$
C. $7.0x10^{-8}$
D. $8.0x10^{-8}$

60. Which one of the following techniques can be used to find exact mass of an isotope?

A. Chromatography
B. Crystallization
C. Mass spectrometry
D. Photo spectrometry

61. When Grignard's reactant reacts with alcohol, what will be the main product along with the side product containing Magnesium?

A. Aldehyde
B. Alkane
C. Alkene
D. Alkyne

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PHYSICS

59. Which one of the following is a base unit?

A. Force
B. Length
C. Pressure
D. Work

60. If two vectors have same magnitudes and directions, they are termed as _____.

A. equal vectors
B. null vectors
C. position vectors
D. unit vectors

61. The shortest distance between any two points is called _____.

A. displacement
B. distance
C. speed
D. velocity

62. How can we see image in a mirror?

A. Due to deflection of light
B. Due to interference of light
C. Due to reflection of light
D. Due to refraction of light

63. Which one of the following is also known as the law of conservation of energy?

A. Carnot's Theorem
B. First Law of Thermodynamics
C. Law of Mass Action
D. Second Law of Thermodynamics

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REHAN AZIZ
21103389961479
Ticket No.

391

QUANTITATIVE REASONING

27. The multiplicative inverse of -5/6 is _____.

A. -5/6
B. -6/5
C. 5/6
D. 6/5

28. 75% of what number is 51?

A. 38.25
B. 68
C. 100
D. 136

29. Cost of 6 books is Rs. 968. The cost price of 3 books is _____.

A. Rs. 161.333
B. Rs. 322.667
C. Rs. 484
D. Rs. 4840

30. 32 - [36 - 2{5 x 2 + 7 - 6}] = _____.

A. -14
B. 2
C. 14
D. 18

31. What is the next term in the sequence 3, 7, 12, 18, 25, 33, _____.

A. 38
B. 39
C. 40
D. 42

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92. If $x = 8^{\frac{4}{3}}$, what is the value of x?

A. $4\frac{3}{4}$
B. 6
C. $10\frac{2}{3}$
D. 12
E. 16

93. What is the value of $m^2 - 4mn + n$ when m=-1 and n=-3.

A. -15
B. -14
C. -11
D. 8
E. 10

94. How many degrees are there in a Pi chart?

A. 90
B. 60
C. 180
D. 360
E. 400

95. The total weight of father and his son is 250 kg. If father's weight is 10 kg more than 3 times his son's weight. What is the weight of the son?

A. 40
B. 50
C. 60
D. 80
E. 90

96. In an examination 200 students failed in English and 800 failed in Physics. If total 900 students failed in any subject then how many failed in both?

A. 30
B. 40
C. 50
D. 100

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N TESTING COU
EDUCATION COMMISSION
NUMBER SLIP
Test

10. The instruction _____ you to switch off electric equipment before you go out.

A. tell
B. tells
C. to tell
D. told

11. Yesterday I went to a party. Now I am going to tell you what I _____ when I arrived.

A. saw
B. see
C. seeing
D. seen

12. We were late and _____ managed to catch the train.

A. fast
B. hardly
C. hurriedly
D. scarcely

13. The goods were never delivered _____ the promise we had received.

A. although
B. because
C. in spite of
D. yet

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51. The property of binding of light around obstacles and spreading of light waves into the geometrical shadow of an obstacle is called _____ of light.

A. diffraction
B. interference
C. reflection
D. refraction

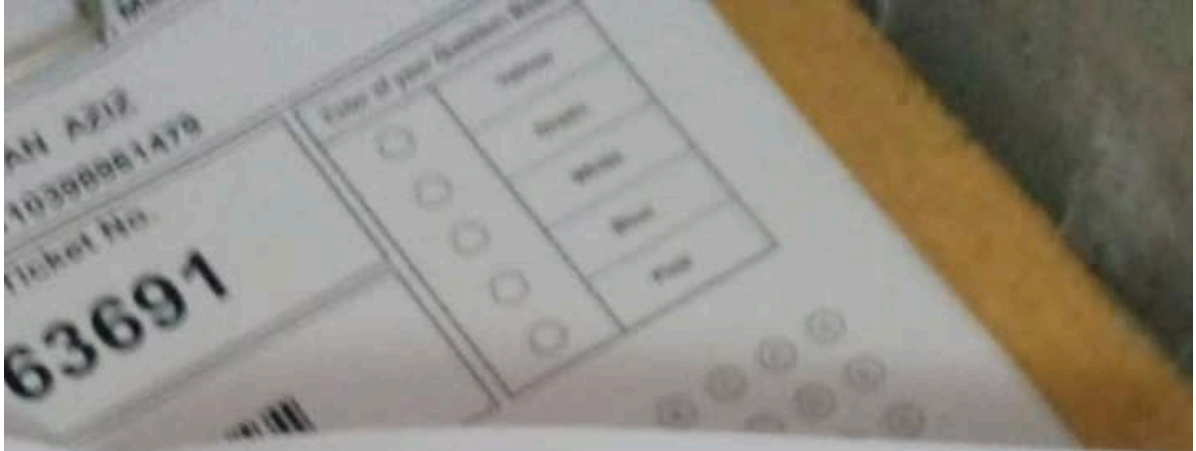
52. "The sum of all the currents flowing towards a point is equal to the sum of all the currents flowing away from the point", is the statement of _____.

A. Flehmening's Rule
B. Kirchhoff's First Rule
C. Kirchhoff's Second Rule
D. Right Hand Rule

53. The shock absorber of a car is an example of _____ oscillation.

A. damped
B. resonance
C. simple hormonic motion
D. undamped

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28. Nadir was able to get his bar of silver back by using his _____.

A. money
B. smile
C. wisdom
D. words

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55. Abdullah gave Rs. 60 to Umar and Umar gave Rs. 90 to Ahmed. After this, all of them have same money. How much more Umar has than Ahmed?

A. Rs. 30
B. Rs. 60
C. Rs. 90
D. Rs. 120

56. The average of first ten whole numbers is _____.

A. 3.5
B. 4.0
C. 4.5
D. 5.5

... in the bank to get a profit of

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39. The mean score of ten students in a test is 49. If one more student is added, then the mean score of students become 50. Find the score of the last student.

A. 10
B. 50
C. 60
D. 62

$\frac{x}{10} = 49$

$x = 490$

11 | 491

40. $3\log_{10}5 + \log_{10}24 - \frac{1}{2}\log_{10}9 =$ _____.

A. 2
B. 3
C. 100
D. 1000

5 | 25

360
130
230

360
180
80

41. Simplify $\frac{\sqrt{25}}{2\sqrt{2}-\sqrt{3}}$.

A. $2\sqrt{2} + \sqrt{3}$
B. $-2\sqrt{2} + \sqrt{3}$
C. $-2\sqrt{2} - \sqrt{3}$
D. $2\sqrt{2} - \sqrt{3}$

$30 \times 8 - \frac{11}{2} \times 20$

$240 - 110$

240
110
130

42. The number of triangles formed inside a polygon is equal to the number of octagons formed inside the polygon. Then the number of sides of polygon are _____.

A. 5
B. 10
C. 11
D. 22

$30 \times 8 - \frac{11}{2} \times 20$

$240 - 110$

$(10-2) \times 180$

$(n-8) \times 180$

240
110
130

43. Ahmed travelled 400 km in 5 hours, then he cycles for 4 hours with a speed of 5 km/h and walks for 1 hour with a speed of 2km/h. What is the average speed of Ahmed during the whole journey?

A. 29 km/h
B. 30 km/h
C. 42.2 km/h
D. 45 km/h

$240 - 110$

240

360
160
400

8
16
24
240

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53. Which of the following is a perfect number?

A. 24
B. 26
C. 28
D. 32

54. The sale price of a book is Rs. 855 with a loss of 5%. What is the cost price of a book?

A. Rs. 870
B. Rs. 900
C. Rs. 910
D. Rs. 993

55. Abdullah gave Rs. 60 to Umar and Umar gave Rs. 90 to Ahmed. After this, all of them have same money. How much more Umar has than Ahmed?

A. Rs. 30
B. Rs. 60
C. Rs. 90
D. Rs. 120

56. The average of first ten whole numbers is _____.

A. 3.5
B. 4.0
C. 4.5
D. 5.5

57. What amount is invested in the bank to get a profit of Rs. 3500 at a rate of 7% for 4 years?

A. 10000
B. 12500
C. 15000
D. 16500

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33. The age of mother is 3 times the age of her daughte
Six years ago, the age of mother was 5 times the ag
her daughter. The age of mother after two years wil
_____.

A. 32 years
B. 36 years
C. 38 years
D. 40 years



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7:32 AM
1.2 K/s
99
50

QUANTITATIVE REASONING

21. A boy cycles for 4 hours with a speed of 6km/h and covers 9km by foot in 7 hours. The average speed of boy is _____.

    A. 3.0 km/h
    B. 3.5 km/h
    C. 7.3 km/h
    D. 8.25 km/h

$S = \frac{15}{11}$ 1.4 ; 11 | 15, 11, 40

22. There were 20 boys and 10 girls in a class. What is the probability of selecting a girl or a boy as a monitor?

    A. 1/30
    B. 1/3
    C. 2/3
    D. 1 ✓

$= \frac{36}{36}$

23. The arithmetic mean of 10 values is 50. If -27 is added to data. Then new arithmetic mean is _____.

    A. 23
    B. 38.5
    C. 43
    D. 47.3

$10 = 50$

$10 - 27 = 50$

24. An amount of Rs. 19000 was invested at a rate of 6% for 4 years on simple interested basis. The total amount in the bank account after 4 years is _____.

    A. Rs. 22000
    B. Rs. 23500
    C. Rs. 23560
    D. Rs. 80560

27
10
17

25. The age of father is 4 times the age of his son. 3 years ago, the age of father was 5 times the age of his son. The present age of father is _____.

    A. 10 years
    B. 24 years
    C. 48 years
    D. 60 years

$F = 4S$

$F - 3 = (S - 3)5$

$4S - 3 = 5S - 15$

26. $[24 - 32 \div \{41 - 25\} - 2 \times 8] + 3 =$ _____.

    A. -3
    B. 0
    C. 6

$[24 - 2 - 2 \times 8] + 3$

41
25

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QUANTITATIVE REASONING

27. Which of the following is a rational number?

A. π
B. 1.252729...
C. 7.323232...
D. $\sqrt{27}$

28. If $f(x) = 3x^4 - 2x^2 + 7$ then value of $f(\sqrt{2})$ is _____

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QUANTITATIVE

21. Number of lines of symmetry that can be passes through regular hexagon is _____.

A. 1
B. 2
C. 3
D. 6

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25. If the cost of 2 copies and 3 books is Rs. 1300 and cost 5 copies and 1 book is 2600. Find the cost of a copy and a book.

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C. Rs. 600
D. Rs. 50000

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C. 21 liters
D. 22 liters

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A. constant
B. identity
C. one-one
D. quadratic

29. The quadratic equation with roots 2, -3 is _____.

A. $x^2 - x + 6$
B. $x^2 + x + 6 = 0$
C. $x^2 + x - 6 = 0$
D. $x^2 - x - 6 = 0$

QUANTITATIVE REASONING

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9. What is/are the charge(s) on Gamma Rays?

A. Negative
B. No charge at all
C. Positive
D. Both positive and negative charges

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16. ***HELMET: HEAD::***

(A) Pendant: Neck

(B) Breastplate: Chest

(C) Pedal: Foot

(D) Knapsack: Back

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QUANTITATIVE REASONING

What is the volume of the cone whose base radius is 3cm and slant height is 5cm?

A. 5π
B. 8π
C. 12π
D. 15π

22. $\frac{27^{3n+1}.81^{-n+2}}{9^{n+5}.3^{3n-1}}$ = _____.

A. 1
B. 3
C. 9
D. 27

23. Car A is moving with speed of 60 km/h and car B is moving with a speed of 80 km/h in the same direction. What is the distance between them after 3 hours if they started at the same point?

A. 20 km
B. 60 km
C. 180 km
D. 240 km

$d = 140 \times t\,(3)$

$= 0$

24. $\{38 - [35 - 2(5 + 7 - 6)]\} \div 3 + 1$ = _____.

A. 4
B. 5
C. 6
D. 32

$= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 9 \times 5\,cm$

$= 130$

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34. The roots of the equation $x^2 + x + 1 = 0$ are _____.
(Where w is a cube root of 1).

A. 1, w, w^2
B. w, w^2
C. 1, w^2
D. 1, w

$x^2 + 1x - x + 1$
$x(x+1) - 1(x+1)$
$(x-1)(x+1)$
$x = 1$

35. Number 5 in 94.60572 is at _____ place.

A. hundredths
B. ones
C. tenths
D. thousandths

30
8
240

36. In a tournament 8 teams were playing. Each team has to play one match with every other team. The total number of matches that were played are _____.

A. 25
B. 28
C. 56
D. 64

$30 \times 8 - \frac{11}{2} \times 20$
$= 240 - 110$
240
110
130

37. A dice is rolled twice. The probability of getting a sum of 6 on dice is _____.

A. 1/36
B. 1/18
C. 5/36
D. 1/6

330
$\frac{n(n+1)}{2} = \frac{n(n-1)}{2}$
130

38. What is the angle between the hands of clock at 20 minutes to 8 O'clock?

A. 5°
B. 10°
C. 15°
D. 20°

$30 - 8 - \frac{11}{2} \times M$
$240 - 110 =$
$\frac{8(7)}{2}$
490
-360
130

$= 30H - \frac{11}{2} \times M$

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2:45 AM 648 B/s
USAT Novemb...

39. The mean score of ten students in a te
more student is added, then the mean
students become 50. Find the score of

A. 10
B. 50
C. 60
D. 62

$\frac{x}{10} = 49$

40. $3\log_{10}5 + \log_{10}24 - \frac{1}{2}\log_{10}9 =$ _____

A. 2
B. 3
C. 100
D. 1000

$5\sqrt{5}$

360
130
230

41. Simplify $\frac{\sqrt{25}}{2\sqrt{2}-\sqrt{3}}$.

A. $2\sqrt{2} + \sqrt{3}$
B. $-2\sqrt{2} + \sqrt{3}$
C. $-2\sqrt{2} - \sqrt{3}$
D. $2\sqrt{2} - \sqrt{3}$

30 x 8 -

240 -

42. The number of triangles formed inside
equal to the number of octagons forme
polygon. Then the number of sides of p
_____.

A. 5
B. 10
C. 11
D. 22

$30 \times 8 - \frac{11}{2} x$

$240 - 110$

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4 hours with a speed of 5 km/ w
with a speed of 2km/h. What is the av
Ahmed during the whole journey?

A. 29 km/h

$240 - 11^0$

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B. 1/3
C. 2/3
D. 1 ✓

$= \frac{36}{36}$

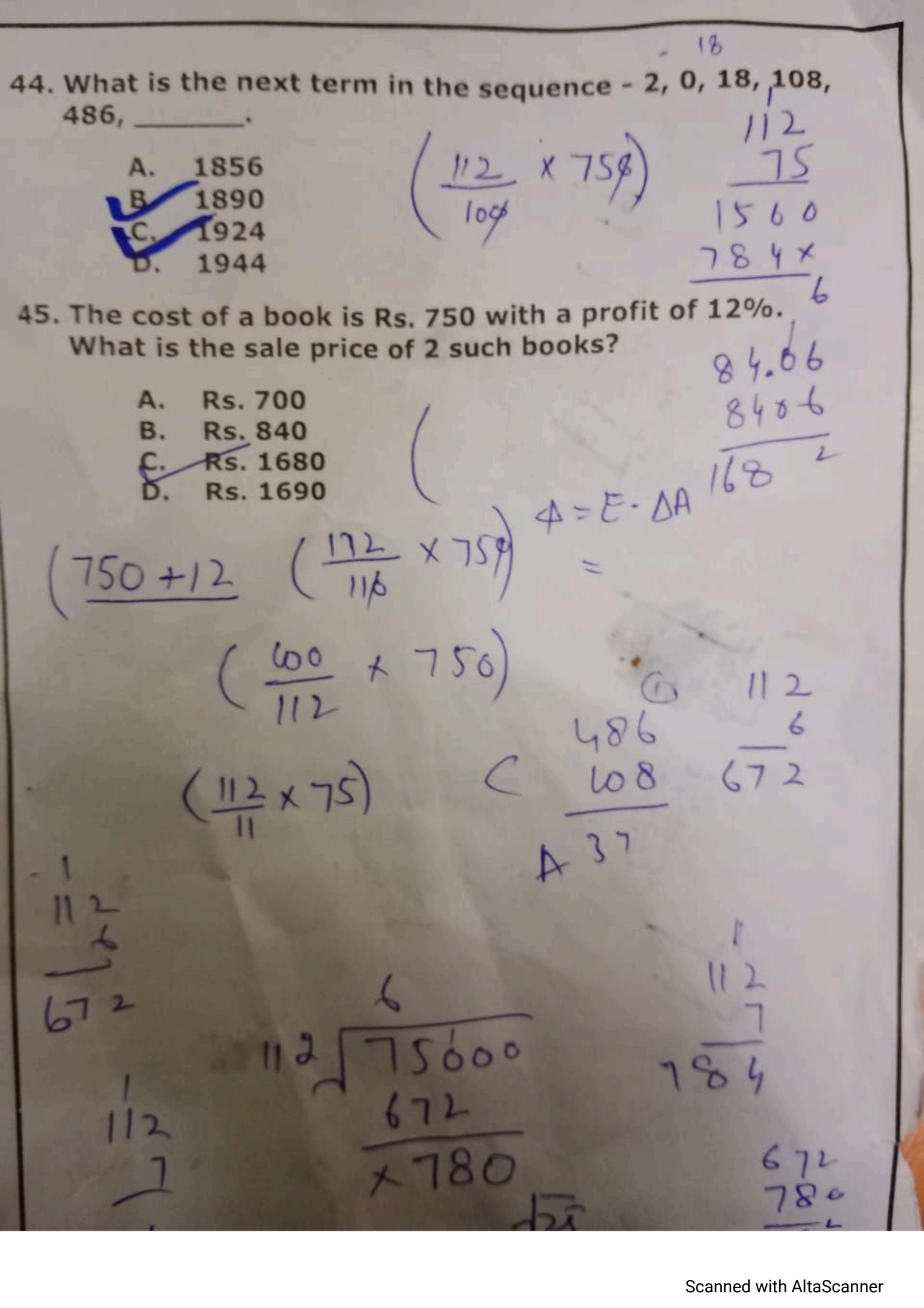
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44. What is the next term in the sequence - 2, 0, 18, 108, 486, _____.

A. 1856
B. 1890
C. 1924
D. 1944

45. The cost of a book is Rs. 750 with a profit of 12%. What is the sale price of 2 such books?

A. Rs. 700
B. Rs. 840
C. Rs. 1680
D. Rs. 1690



6. I have _____ finished my work.

A. as
B. just
C. since
D. yet

7. Which story would you like me _____ today?

A. tell
B. telling
C. to tell
D. told

8. Tomorrow morning by this time we will _____ in an airplane.

A. flying
B. flew
C. fly
D. have flown

Four lettered pairs (A to D) follow a related pair of words given in the question. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair in capital letters.

9. BIRD: BEAK::

A. bee: hive
B. horse: hoof
C. man: mouth
D. snake: poison

10. MOUNTAIN: PEAK::

A. desert: sand
B. river: bank
C. sea: waves
D. tree: top

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43. Which of the following values of heat of formation indicates that the product is least stable?

A. -94 k cal
B. -231 k cal
C. +21.4 k cal
D. +64.8 k cal

44. The enthalpy of certain reaction at 273 K is -20.75 kJ. The enthalpy of same reaction at 373 K (if heat capacities of reactants and products is same) will be:

A. -20.75 kJ
B. -2075 kJ
C. Zero
D. $-20.75 \times \frac{373}{273}$ kJ

45. Most abundant salt of sodium in nature is:

A. $NaNO_3$
B. Na_2SO_4
C. NaOH
D. NaCl

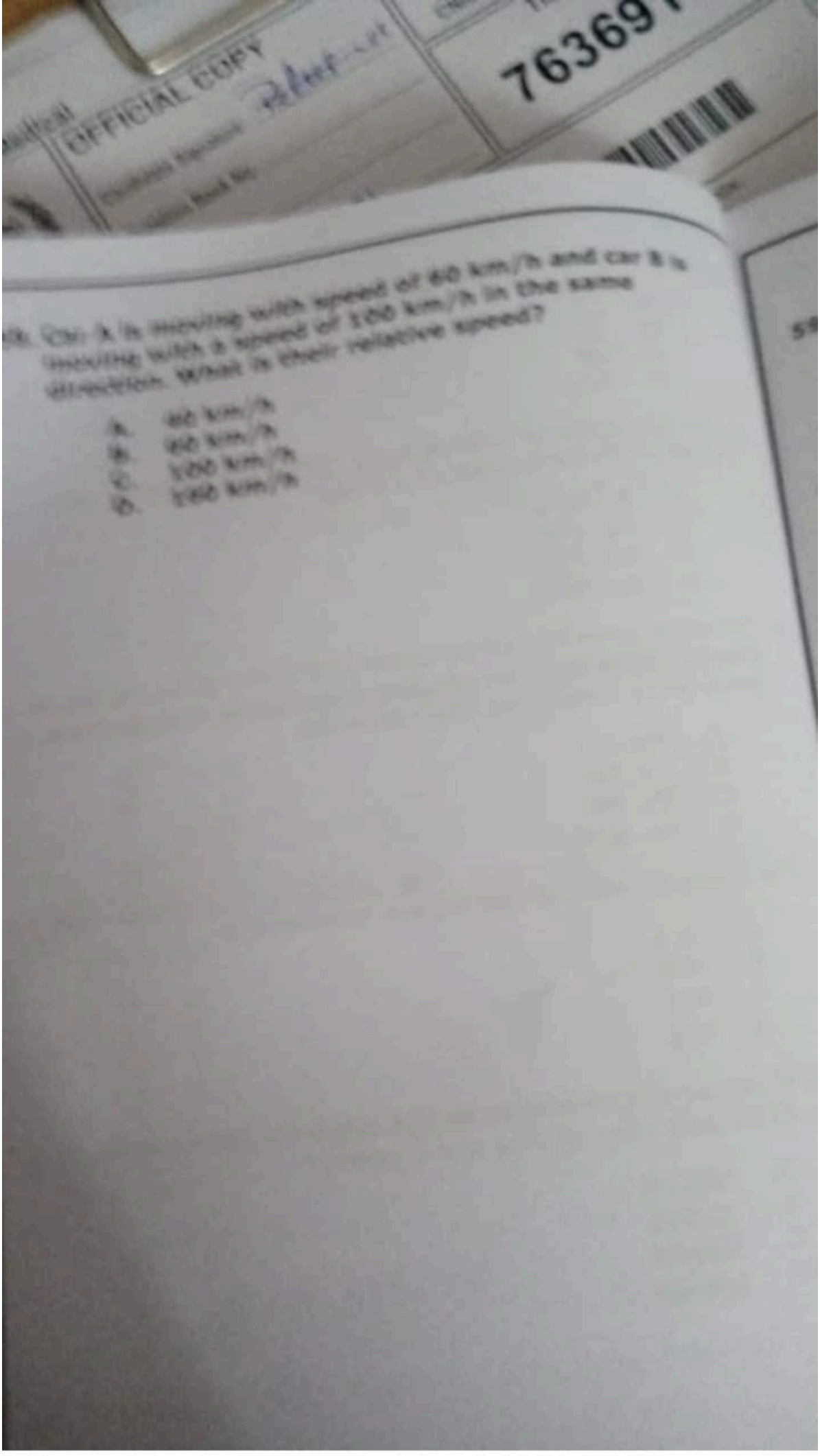
46. A gas that reacts with CaO and not with $NaHCO_3$ is:

A. CO_2
B. Cl_2
C. O_2
D. N_2

47. Which of the following statements is correct?

A. H_3PO_3 is dibasic and reducing
B. H_3PO_3 is tribasic and reducing
C. H_3PO_3 is tribasic and non-reducing
D. H_3PO_3 is dibasic and non-reducing

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D. 1030 Watt

53. The SI unit of magnetic flux is Weber. One Weber can also be expressed as _____.

A. Joule/Ampere
B. Joule/meter
C. Newton/meter
D. Tesla/m^2

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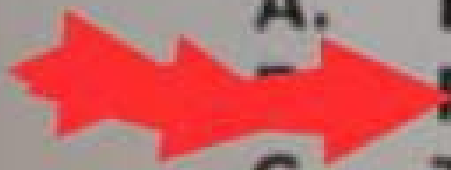
6. OCEAN: WEEDS::

A. kids: parents
B. movie: actors
C. loyal: state
D. forest: bushes

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82. Which one of the following is NOT abiotic factor?

A. Food
B. People
C. Temperature
D. Water



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PHYSICS

46. What is the SI units of electric flux?

A. Nm^2C^{-1}
B. NmC^{-1}
C. Nm^2C^{-2}
D. Nm^2C^{-3}

47. What are the base units of pressure?

A. kgm^2s^{-2}
B. $kgms^{-2}$
C. kgm^2s^{-3}
D. $kgm^{-1}s^{-2}$

48. Plastic and synthetic rubber are examples of which type of solids?

A. Amorphous solids
B. Crystalloid solids
C. Glassy solids
D. Polymeric solids

49. Which type of interaction took place when very high energy photons such as gamma rays interact with matter?

A. Annihilation of matter
B. Compton effect
C. Pair production
D. Photoelectric effect

50. Which one of the following is a non-renewable source of energy?

A. Coal
B. Sunlight
C. Tide
D. Wind

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64. Which one of the following is the SI unit of work?

A. Coulomb
B. Joule
C. Meter
D. Newton

65. A modified form of galvanometer used to measure the resistance is termed as _____.

A. ammeter
B. ohmmeter
C. voltmeter
D. Wheatstone bridge

66. Which of the following are the materials in which valance electrons are free and they can conduct electricity at normal temperatures?

A. Conductors
B. Insulators
C. Semi-conductors
D. Super conductor

67. In electronic circuits, when there are two or more inputs for a single output, and all of them are necessary for output, the Gate used is called _____.

A. AND Gate
B. NAND Gate
C. NOR Gate
D. OR Gate

68. The mass of the body can be converted into energy by expression _____.

A. $E=mc^2$
B. $E=m/c^2$
C. $E=m^c$
D. $E=m^2c$

69. Wha

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70. What will be the molecular mass of NaOH?

A. 1
B. 16
C. 23
D. 40

71. Pick the process used to separate mixture of inks on a blotting paper?

A. Chromatography
B. Crystallization
C. Filtration
D. Sublimation

72. Which one of the following law is related to the diffusion of gases?

A. Avogadro's Law
B. Boyle's Law
C. Charles's Law
D. Gram's Law

73. Conversion of liquid water into an ice in refrigerators is termed as _____.

A. boiling
B. evaporation
C. freezing
D. sublimation

74. Which of the following theories explain the shapes of molecules?

A. Lewis Model
B. Molecular orbital theory
C. Valance bond theory
D. Valence shell electron pair repulsion theory

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25. Pipe A can fill a tank in 2 hours, pipe B can fill the ta in 1 hour 30 minutes, a pipe C can fill it in 2 hours 3 minutes and pipe D can fill in 1 hour. What is the le time in which tank will be full only when two pipes a opened?

A. 18 minutes
B. 36 minutes
C. 54 minutes
D. 150 minutes

26. Ahmed and Fahad together have Rs. 1210. If 6/15 of Ahmed's amount is equal to 3/5 of Fahad's amount, how much amount does Ahmed have?

A. Rs. 450
B. Rs. 484
C. Rs. 720
D. Rs. 726

27. The sum of, sum and product of roots of the equation $3x^2 + 7x - 1 = 0$ is _____.

A. -8/3
B. -2
C. 2
D. 8/3

28. The sum of two numbers is 25 and their difference is 75 the ratio of two numbers is _____.

A. -1:2
B. 1:2
C. 1:3
D. 13:6

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Four lettered pairs (A to D) follow a related pair of words given in the question. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair in capital letters.

16. FLOWER: BUNCH::

A. captain: team
B. man: men
C. manager: business
D. ship: fleet

17. BLACKSMITH: IRON::

A. cook: kitchen
B. farmer: grower
C. potter: clay
D. shoemaker: cobbler

COW: CALF::

A. dog: bark
B. fish: water
C. horse: foal
D. tortoise: shell

ISE: FOOLISH::

A. bold: brave
B. free: fair
C. simple: easy
D. single: many

20. COMPUTER: MONITOR::

A. bicycle: wheel
B. pencil: eraser
C. purse: money
D. road: traffic

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"A most terrible thing happened at the riverbank," he answered softly. "A hawk suddenly swooped down and carried off your son. I could do nothing to help him." What a liar you are! How could a hawk carry off a boy as big as my son? Laeeq said angrily.

Nadir looked at his friend straight in the eyes. "Well, my dear friend, Laeeq," he said calmly, "if mice can eat silver, then hawks can carry off boys."

For a while Laeeq said nothing. Then he knew that Nadir had tricked him just as he had tricked Nadir. So, he said, "Give me back my son and I will give you back your bar of silver."

22. Nadir had to travel abroad in order to _____

A. earn some money
B. get a bar of silver
C. meet his relatives
D. visit a holy place

23. Nadir left his bar of silver with Laeeq because he _____ him.

A. believed
B. feared
C. loved
D. trusted

24. When Laeeq refused to return his bar of silver Nadir did NOT become _____.

A. afraid
B. angry
C. silent
D. tired

25. Laeeq was a _____ man.

A. caring
B. dishonest
C. poor
D. stupid

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54. Which one of the following chemical products is being used for refining of sugar and other food products?

A. Brine solution
B. Gypsum
C. Lime
D. Nitric acid

55. Sodium chloride is NOT soluble in _____.

A. Acetic acid
B. Dimethyl ether
C. Ethanol
D. Water

56. Sodium dichromate can be converted into potassium dichromate by reacting it with _____.

A. NaCl
B. NaBr
C. KCl
D. $MgCl_2$

57. Which one of the following are network covalent type of solids with very strong covalent bond?

A. Diamond
B. Magnesium oxide
C. Rhombic Sulphur
D. Sodium

58. Which product will be released at anode if we electrolyzed aquas solution of H_2SO_4?

A. H_2
B. O_2
C. S_8
D. SO_2

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22. There were 20 boys and 10 girls in a class. What is the probability of selecting a girl or a boy as a monitor?

A. 1/30
B. 1/3
C. 2/3
D. 1 ✓

$= \frac{36}{36}$

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QUANTITATIVE REASONING

21. The value x in the linear equation $\frac{3x}{2} \quad \frac{x-2}{3} \quad \frac{}{2}$

_____.

A. -3
B. -5
C. 3
D. 5

22. Pipe A can fill a tank in 30 minutes and
the tank in 50 minutes. In how much
full if both pipes open together?

A. 18 minutes
B. 18 minutes 45 seconds
C. 30 minutes
D. 80 minutes

23. The sale price of a books is Rs. 630 with
What is the cost price of book?

A. Rs. 595
B. Rs. 600
C. Rs. 610
D. Rs. 661

24. If $f(x) = x^2 + 5kx + 70$ and $f(-3) = 4$ then

_____.

A. -5
B. -47/5
C. 5
D. 47/5

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33. A coin is tossed twice. The probability of getting doublets is _____.

A. 1/4
B. 1/2
C. 3/4
D. 1

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23. What is the next term in the sequence 1, 2, 4, 7, 11, 16, 22, 29?

A. 37
B. 39
C. 41
D. 57

24. What is the angle between the hands of a clock at 6 O'clock?

A. 90°
B. 120°
C. 150°
D. 180°

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78. Sum of interior angles of octagon is _____.

A. 540°
B. 720°
C. 900°
D. 1080°

79. 60% of what number is 270?

A. 200/9
B. 162
C. 450
D. 475

80. In 7098, the sum of place and face value of 7 is _____.

A. 14
B. 77
C. 7007
D. 14000

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C. 30 cm

D. 40 cm

11. $\omega^{88} = ?$

A. 1

B. ω^2

C. ω

D. 0

12. $\omega^n = ?$, when $n = 3k$

A. 0

B. ω

C. 1

D. $\frac{1}{\omega}$

13. The number of real roots in cube roots of 8 is?

B. One

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25. Pipe A can fill a tank in 2 hours, pipe B can fill the ta
in 1 hour 30 minutes, a pipe C can fill it in 2 hours ?
minutes and pipe D can fill in 1 hour. What is the le.
time in which tank will be full only when two pipes a
opened?

2 hr

A. 18 minutes ✓ 60 1hr. 30
B. 36 minutes ✓ 60
C. 54 minutes ✓ 120 2hr 30
D. 150 minutes ✓ 1

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QUANTITATIVE REASONING

21. The value x in the linear equation $\frac{3x}{2} - \frac{x-2}{3} = \frac{13}{2}$

_____.

A. -3
B. -5
C. 3
D. 5

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88. A factory worker can produ
If there are 42 workers in t
produced every hour?

A. 147 toys
B. 1176 toys
C. 200 toys
D. 150 toys
E. 400 toys

89. If $p - 7 = 5 - p$, then $p =$

A. -6
B. -2
C. 1
D. 6
E. 12

90. Muzamil's quiz scores in ma
60. What was the average o

A. 80.2
B. 81
C. 81.5
D. 82
E. 84

91. $(a - b)^2 =$

A. $a^2 - b^2$
B. $a^2 + b^2$

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51. The measurement of wavelength and intensities of electromagnetic radiation emitted or absorbed by atom is known as _____ spectra.

A. atomic
B. light
C. molecular
D. solar

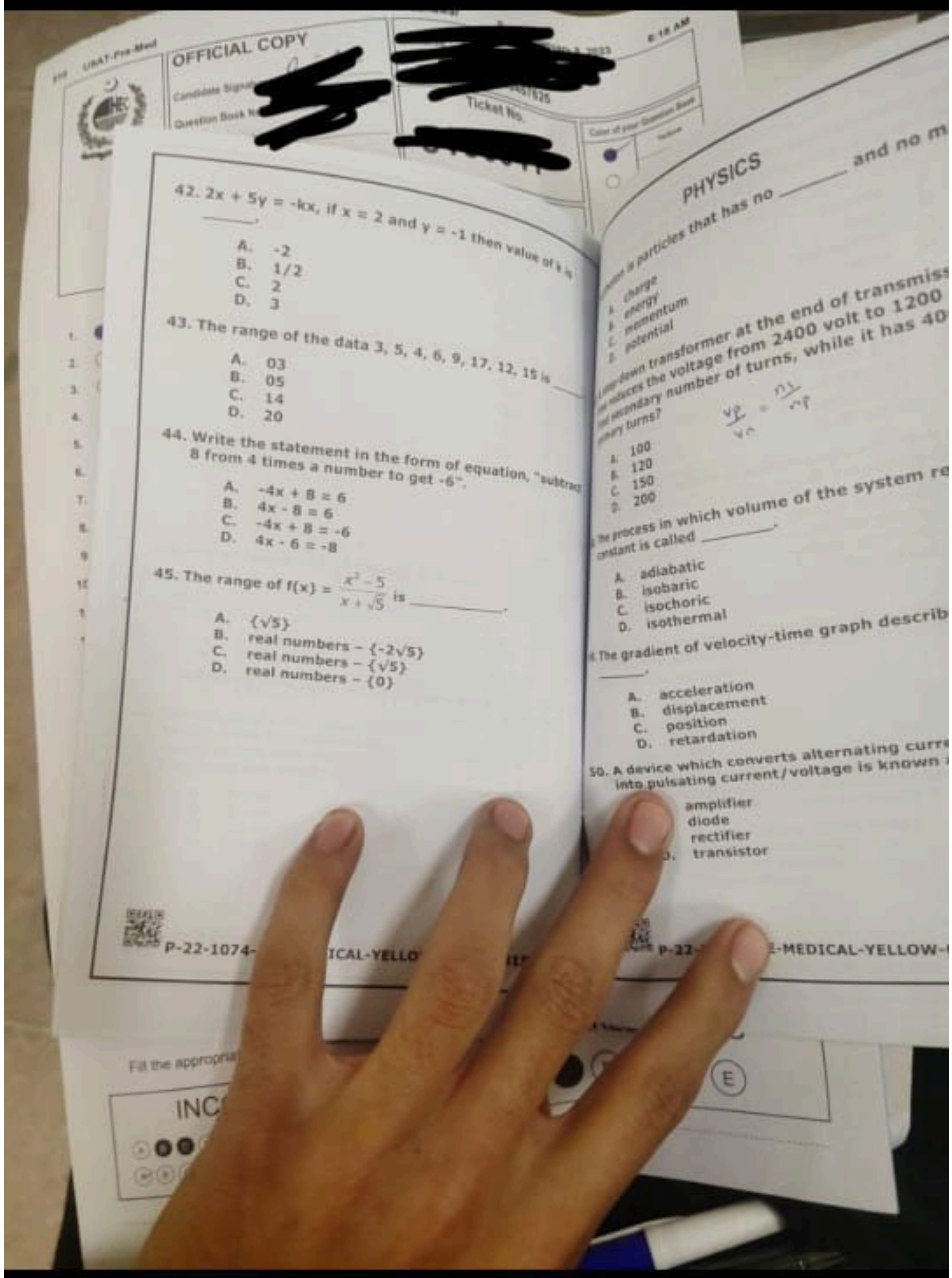
52. A 70 Kg boy runs up hill through height of 3m in 2s, what will be average power?

A. 850 Watt
B. 930 Watt
C. 1010 Watt
D. 1030 Watt

53. The SI unit of magnetic flux is Weber. One Weber can also be expressed as _____.

A. Joule/Ampere
B. Joule/meter
C. Newton/meter
D. Tesla/m^2

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QUANTITATIVE REASONING

21. A boy cycles for 4 hours with a speed of 6km/h and covers 9km by foot in 7 hours. The average speed of boy is _____.

    A. 3.0 km/h
    B. 3.5 km/h
    C. 7.3 km/h
    D. 8.25 km/h

$S = \frac{15}{11}$ 1.4 / 11) 15 / 11 / 40

22. There were 20 boys and 10 girls in a class. What is the probability of selecting a girl or a boy as a monitor?

    A. 1/30
    B. 1/3
    C. 2/3
    D. 1 ✓

$= \frac{30}{30}$

23. The arithmetic mean of 10 values is 50. If -27 is added to data. Then new arithmetic mean is _____.

    A. 23
    B. 38.5
    C. 43
    D. 47.3

10 = 50

10 − 27 = 50

24. An amount of Rs. 19000 was invested at a rate of 6% for 4 years on simple interested basis. The total amount in the bank account after 4 years is _____.

    A. Rs. 22000
    B. Rs. 23500
    C. Rs. 23560
    D. Rs. 80560

27 / 10 / 17

25. The age of father is 4 times the age of his son. 3 years ago, the age of father was 5 times the age of his son. The present age of father is _____.

    A. 10 years
    B. 24 years
    C. 48 years
    D. 60 years

F = 4S

F − 3 = (S − 3)5

4S − 3 = 5S − 15

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5. SINGULAR: PLURAL::

A. below: beneath
B. haughty: vain
C. least: most
D. lift: soar

6. OCEAN: WEEDS::

A. kids: parents
B. movie: actors
C. loyal: state
D. forest: bushes

7. FINGER: GUITAR::

A. beak: bird
B. level: vertical
C. snake: mice
D. stick: drum

8. TEETH: BITE::

A. knife: cut
B. tree: green
C. drop: ocean
D. language: alphabet

9. SUMMARY: PRECIS::

A. ladle: pan
B. monitor: screen
C. order: obedience
D. reverse: push

Questions 10-15

After years of

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C. 720^0
D. 1080^0

39. The amplitude of function $6\sin(3x + 5)$ is _____.

A. -5/3
B. -5
C. 06
D. 2n

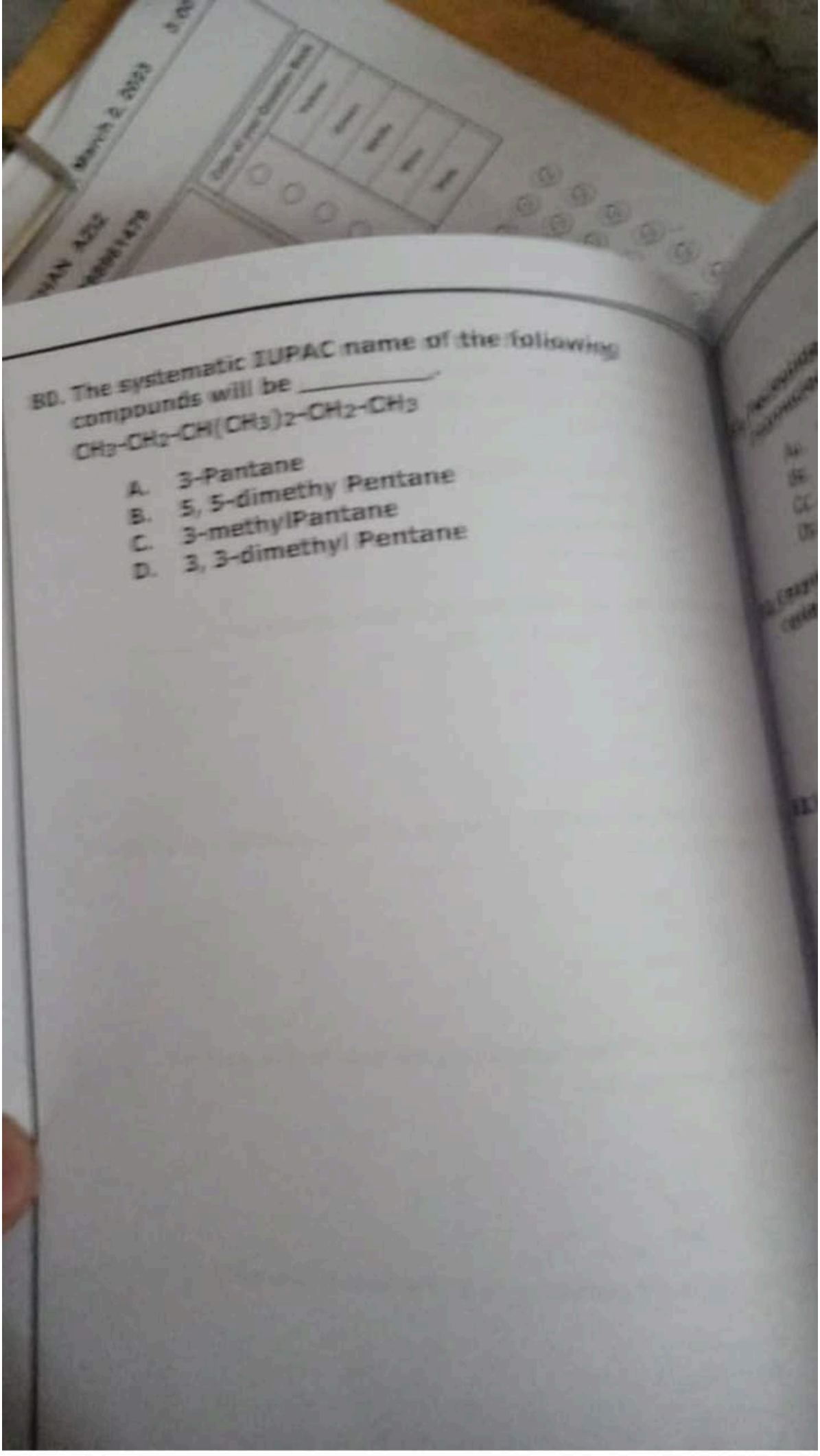
40. If $x^3 - 2x^2 - 5x + 16$ is divided by $2x - 6$ then remainder is _____.

A. -14
B. 0
C. 10
D. 15

41. The sum of the place and face value of 6 in number 54689 is _____.

A. 0
B. 594
C. 600
D. 606

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80. The systematic IUPAC name of the following compounds will be _____.
CH_3-CH_2-$CH(CH_3)_2$-CH_2-CH_3

A. 3-Pantane
B. 5, 5-dimethy Pentane
C. 3-methylPantane
D. 3, 3-dimethyl Pentane

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13. In a tournament 8 teams were playing. Each team has to play a match with every other team. Two semifinals and one final are played. How many total matches were played?

A. 25
B. 28
A. 31
B. 34

SAMPLE PAPER

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48. Pipe A can fill a tank in 2 hours, Pipe B can fill the tank in 1 hour 30 minutes and a Pipe C can fill it in 2 hours 30 minutes. Which pipe(s) is/are more efficient?

A. Pipe A
B. Pipe B
C. Pipe C
D. Both A and C

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11. Ahmed has 95/3 liters of paint. After using 59/12 liters in room and 37/12 liters in kitchen, how much paint is left?

A. 7/12 liters
B. 5/3 liters
C. 71/6 liters
D. 71/3 liters

12. Sum of roots of $9x^2 + 21x - 8 = 0$ is _____.

A. -7/3
B. -8/9
C. 8/9
D. 7/3

T-PRE-MEDICAL

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44. Pipe A can fill a tank in 45 minutes. If both pipes A and B are opened the tank is full in 15 minutes. In how much time tank will be full if pipe B is opened?

A. 20 minutes
B. 22.5 minutes
C. 30 minutes
D. 35.5 minutes

45.

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A. 360°
B. 600°
C. 720°
D. 1080°

39. The amplitude of function 6sin(3x + 5) is _____

A. -5/3
B. -5
C. 06
D. 2n

40. If $x^3 - 2x^2 - 5x + 16$ is divided by 2x - 6 then r is _____.

TECNO CAMON
AI TRIPLE CAMERA

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14. The angle between the hands of a clock at 4:30 is _____.

A. 30°
B. 35.5^{o}
C. 37.5^{o}
D. 40°

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22. Pipe A can fill a tank in 30 minutes and [illegible] the tank in 50 minutes. In how much [illegible] full if both pipes open together?

A. 18 minutes
B. 18 minutes 45 seconds
C. 30 minutes
D. 80 minutes

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36. If $a + b = 10$ and $a - b = 6$, then the value of ab is ____

A. 16
B. 32
C. 48
D. 64

37. The value of m in the ratio $3: m - 7 = 9:12$ is ____

A. -11
B. -3
C. 3
D. 11

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29. The sale price of an article is Rs. 3597 with a profit o
9%. The cost price of article is _____.

A. Rs. 310.50
B. Rs. 3197.33
C. Rs. 3250
D. Rs. 3300

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Choose the lettered word or phrase that is
NEARLY SIMILAR in meaning to the word
capital letters.

6. The AUDIENCES are _____ of a performance.

A. enemies
B. organizers
C. players
D. viewers

7. If something DEVELOPS it _____.

A. falls
B. grows
C. performs
D. stops

Choose the lettered word or phrase that is
NEARLY OPPOSITE in meaning to the word
capital letters.

8. He is a SHREWD fellow and not _____.

A. cooperative
B. cruel
C. dull
D. honest

9. DOWDY atmosphere is not _____.

A. calm
B. cheerful
C. dangerous
D. healthy

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36. If $a + b = 10$ and $a - b = 6$, then the value of ab is ____

A. 16
B. 32
C. 48
D. 64

37. The value of m in the ratio $3: m - 7 = 9:12$ is ____

A. -11
B. -3
C. 3
D. 11

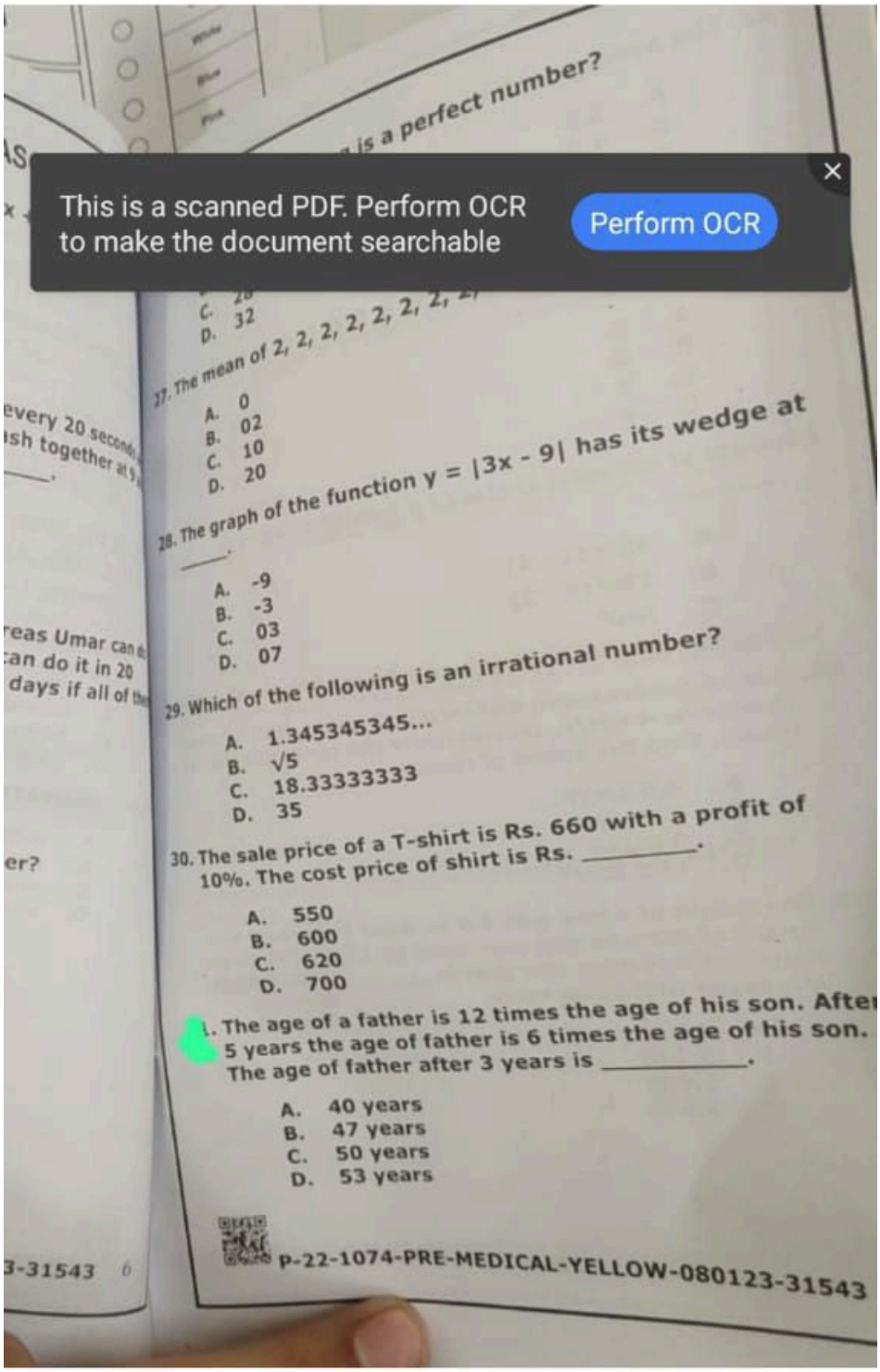
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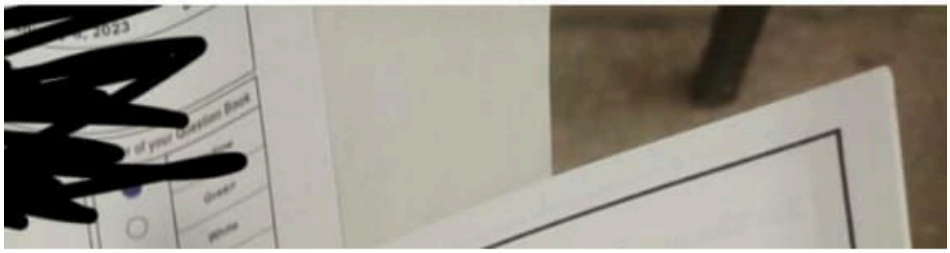
D. 5.5

57. What amount is invested in the bank to get a profit of Rs. 3500 at a rate of 7% for 4 years?

A. 10000
B. 12500
C. 15000
D. 16500

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D. 11

34. The cost price of 8 suits is Rs. 13400. What is the cost price of such 5 suits?

A. Rs. 8000
B. Rs. 8275
C. Rs. 8375
D. Rs. 8425

35. The sale price of an article is Rs. 3597 with a profit of 9%. The cost price of article is _____.

A. Rs. 310.50
B. Rs. 3197.33
C. Rs. 3250
D. Rs. 3300

is 45 minutes. If both pipes A and

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D. 22 liters

28. If f(x) = 5a then it is called _____ function.

A. constant
B. identity
C. one-one
D. quadratic

-3 is _____

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(C) 81.2

(E) 812

13. Ali and Omer share an apartment. If each month Ali pays x dollars and Omer pays y dollars, what percent of the total cost does Ali pay?

(A) $(x + y)100$

(B) $\frac{x}{y}\%$

(C) $\frac{100x}{y}\%$

(D) $\frac{100x}{x+y}\%$

(E) $\frac{100}{xy}\%$

14. There are 240 seats in the balcony of a theater. The number of seats in each row is 14 more than the number of rows. Find the number of rows.

(A) 6

(B) 8

(C) 10

(D) 12

(E) 24

15. Which of the following cannot be expressed as the sum of three consecutive integers?

(A) 27

(B) 26

(C) 21

(D) 42

(E) 39

16. 6 men can do a piece of work in 12 days. How many men are needed to do the work in 18 days?

(A) 2 men

(B) 6 men

(C) 4 men

(D) 3 men

(E) 5 men

17. What is the value of x if $3^{x+1} = 243$?

(A) 3

(B) 5

(C) 7

(D) 4

(E) 2

18. A lighthouse blinks regularly 5 times

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D. -1

9. Two natural numbers whose sum is 25 and difference is 5, are
   A. 25, 20
   B. 20, 10
   C. 20, 5
   D. 15, 10

10. The length of rectangle is twice as much as its breadth. If the perimeter is 120 cm, the length of the rectangle is
   A. 10 cm
   B. 20 cm

98

was 3 time as old as his son. How old is the son now?
   A. 18 years
   B. 36 years
   C. 8 years
   D. 16 years

15. The two consecutive positive integers whose product is 56 are
   A. 7, 8
   B. 14, 4

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If $\tan A = 5/4$ then $\cot B =$ _____.

A. 4/5
B. 5/4
C. 4
D. 5

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12. **A garrison has provision for 12 days for 1000 men. At the end of 2 days 200 more men were admitted. How long will the food now last?**

A. 8 days **B. 8.33 days** **C.9 days** **D.9.33 days**

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B. -3
C. 03
D. 07

29. Which of the following is an irrational number?

A. 1.345345345...
B. $\sqrt{5}$
C. 18.33333333
D. 35

660 with a profit of

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30. If $f(x) = x^2 + 5kx - 70$ and $f(-3) = -1$ then value of k is _____.

A. -62/15
B. -4
C. 4
D. 62/15

$-1 = 9 - 15 - 70$

$= 9 - 85$

-85
9
$\overline{76}$

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QUANTITATIVE REASONING

76. $\frac{3}{5} + \frac{6}{5} \div \frac{7}{15} + \frac{3}{20} = \frac{5 \div 7}{6 \div 15} + \frac{3}{20}$

A. -49/60
B. -7/60
C. 11/60
D. 49/60

77. In a football league, 9 teams were playing. How many matches were played if every team had to play one match with every other team?

A. 20
B. 36
C. 72
D. 81

78. Sum of interior angles of octagon is _____.

A. 540°
B. 720°
C. 900°
D. 1080°

79. 60% of what number is 270?

A. 200/9
B. 162
C. 450

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37. A dice is rolled twice. The probability of getting a sum of 6 on dice is _____

A. 1/36
B. 1/18
C. 5/36
D. 1/6

360
330

$\frac{n(n+1)}{2} = \frac{n(n-1)}{2}$

130

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B. 2/5
C. 5/3
D. 5

If tanA = 5/4 then cotB = _____.

A. 4/5
B. 5/4
C. 4
D. 5



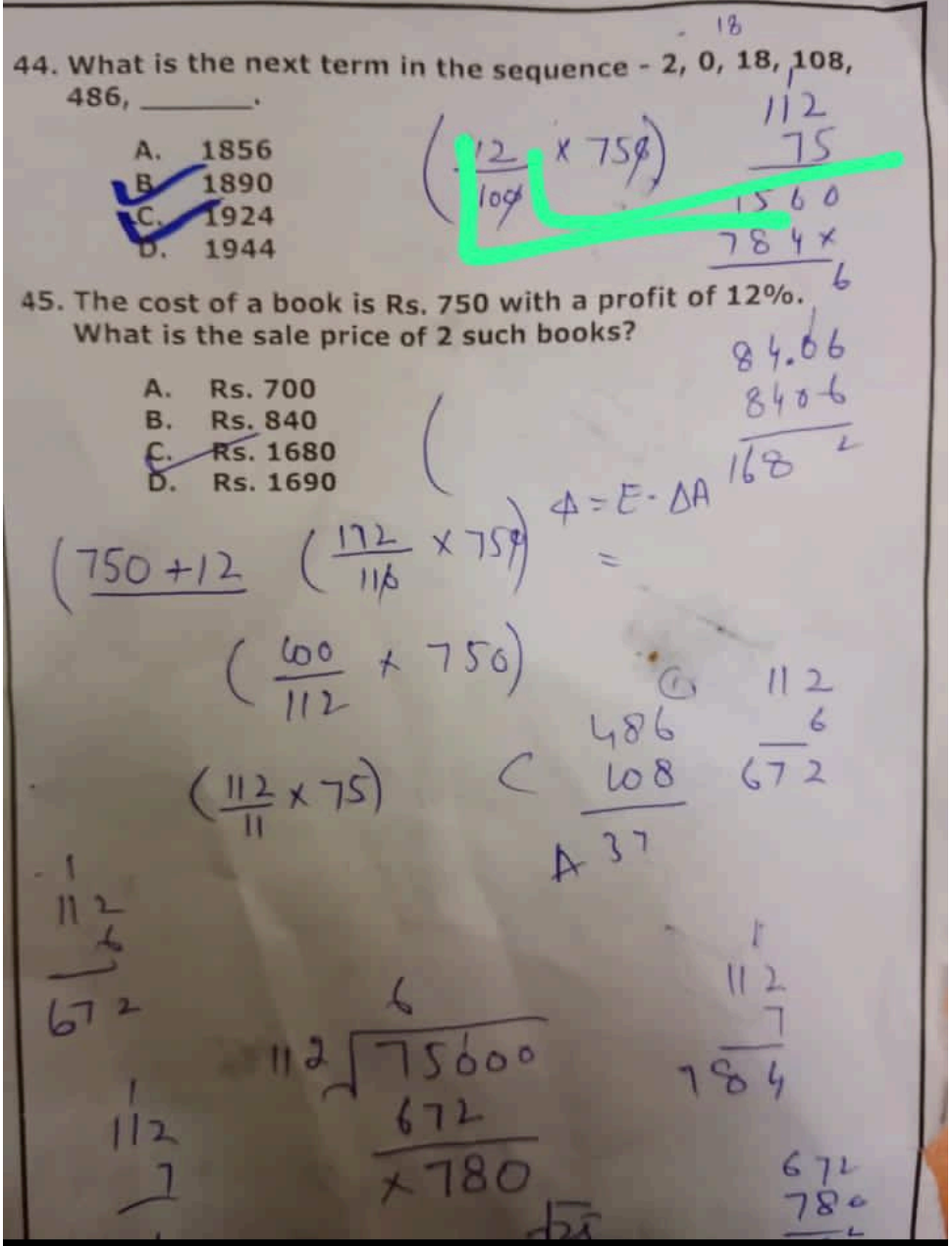
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44. What is the next term in the sequence - 2, 0, 18, 108, 486, _____.

A. 1856
B. 1890
C. 1924
D. 1944

45. The cost of a book is Rs. 750 with a profit of 12%. What is the sale price of 2 such books?

A. Rs. 700
B. Rs. 840
C. Rs. 1680
D. Rs. 1690



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87. If $5x + 7y = 17$ and $3x + 9y = 11$ then value of $x + 2y$ = _____.

A. -7
B. -7/2
C. 7/2
D. 7

$5x + 7y = 17$
$-3x - 9y = -11$
$-2x - 2y = 6$

88. The average of supplementary angles is _____.

A. 0°
B. 45°
C. 90°
D. 180°

89. The sum of first 20 natural numbers is _____.

A. 190
B. 200
C. 210
D. 220

$S = a(1-r^n)/(1-r)$

90. Car A is moving with a speed of 110 km/h in the opposite direction of car B, which is moving with a speed of 50 km/h. What is their relative speed?

A. 50 km/h
B. 60 km/h
C. 110 km/h
D. 160 km/h

91. The sale price of shirt is Rs 736 with a loss of 8%. The cost price of shirt is _____

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A. 16
B. 32
C. 48
D. 64

37. The value of m in the ratio 3: m - 7 = 9:12 is _____

A. -11
B. -3
C. 3
D. 11

38. The linear equation 3x = 10 represents a _____ line

A. horizontal
B. oblique
C. vertical
D. y-axis

39. If A:B = 5:7 and C:B = 5:7 then A:B:C = _____

A. 5:7:5
B. 7:5:7
C. 25:35:40
D. 49:35:25

40. 38 - [36 - 2(5 - 7 + 6)] ÷ 4 = _____

A. 5/2
B. 7
C. 31
D. 32

41. If the cost of 2 hens and 3 goats is Rs. 47000 and of 5 hens and 1 goat is Rs. 20000. What is the cost of 1 hen?

A. Rs. 1000
B. Rs. 15000
C. Rs. 16000
D. Rs. 20000

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87. If $5x + 7y = 17$ and $3x + 9y = 11$ then value of $x + 2y$ = _____.

A. -7
B. -7/2
C. 7/2
D. 7

$5x + 7y = 17$

$-\ 3x - 9y = -11$

$-2x - 2y = 6$

$-2(x + y)$

$= -6$

$= -3$

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D. 150 minutes

26. Ahmed and Fahad together have Rs. 1210. If 6/15 of Ahmed's amount is equal to 3/5 of Fahad's amount, how much amount does Ahmed have?

A. Rs. 450
B. Rs. 484
C. Rs. 720
D. Rs. 726

245
5 | 1210
10 21
0 20

$= \frac{3}{5} \times 1210 \times \frac{1}{6}$ 245

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92. A man covers some distance in 5 hours. If he moves faster by 4 km/h, he will cover the same distance in 3 hours. Find the distance covered by the man.

A. 15 km
B. 24 km
C. 30 km
D. 36 km

93. Value of the determinant of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is _____.

A. -2
B. 0
C. 1
D. 2

94. The sum of two-digits of a number is 9. If the digits are interchanged the new number so formed is 45 greater than the original number. The original number is _____.

A. 27
B. 36
C. 45
D. 72

95. Sum of roots of the equation $7x^2 - 13x - 40 = 0$ is _____.

A. -40/7
B. -13/7
C. 13/7
D. 40/7

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A. 10 years
B. 24 years
C. 48 years
D. 60 years

F = 4S
F-3 = (S-3)5
4S-3 = 5S-15

26. $[24 - 32 \div \{41 - 25\} - 2 \times 8] + 3 =$ _____.

A. -3
B. 0
C. 6
D. 9

$[24 - 2 - 2 \times 8] + 3$
$[24 - 2 - 18] + 3$

27. Calculate the area of a right triangle, whose base length is 5cm and length of hypotenuse is 13cm.

A. 30 cm^2
B. 32.5 cm^2
C. 60 cm^2
D. 65 cm^2

$[4 - 20] + 3$
$[4 + 3] = 7$

28. If $m + n + p = 8$ and $m^2 + n^2 + p^2 = 20$, then the value of $mn + np + mp =$ _____.

A. -12
B. 12
C. 22
D. 44

29. If $(x + 2)$ is a factor of $3x^2 - 4kx - 4k^2$, then the value of k is _____.

A. -3
B. -1
C. 1
D. 2

$12 + 8R - 4k^2 =$
$12 + 2(4R - 2k)$

3 / 4

30. Solution of the equation $\sqrt{5x-6} - \sqrt{x+10} = 0$ is _____.

A. -4
B. 0
C. 4
D. 8

$\sqrt{5x-6} = \sqrt{x+10}$
$5x - 6 = x + 10$
$4x = 16$
$x = 4$

31. To increase the tourism the government has to construct a road of 86 kms long. 39 kms of the road is

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16. Factors of $x^3 + 4x^2 - 21x = 0$ are _____

A. -7, -3, 0
B. -7, 3, 0
C. -3, 7, 0
D. 3, 7, 0

17. Two dice are rolled. What is the probability of getting same faces on top?

A. 5/36
B. 1/6
C. 7/36
D. 1

18. Ahmed added two containers of paint containing 40 liters and 55 liters of paint. After using 49.3 liters in room and 34.7 liters in kitchen, how much paint is left

A. 10 liters
B. 11 liters
C. 12 liters
D. 21 liters

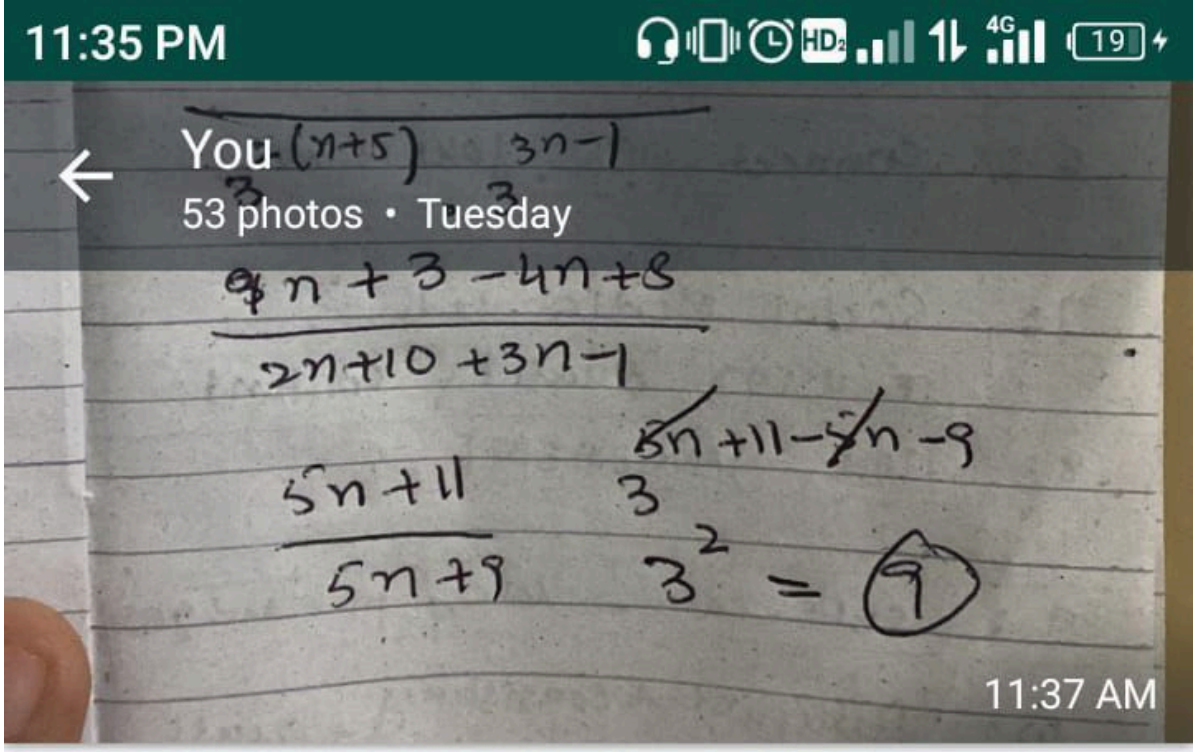
19. If $f(x) = 5x$ then it is called _____ function.

A. constant
B. identity
C. one-one
D. quadratic

20. The sale price of a book is Rs. 856 with a profit of 7% What is the cost price of book?

A. Rs. 795
B. Rs. 796.08
C. Rs. 800
D. Rs. 805

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34. The roots of the equation $x^2 + x + 1 = 0$ are _____.
(Where w is a cube root of 1).

A. 1, w, w^2
B. w, w^2
C. 1, w^2
D. 1, w

35. Number 5 in 94.60572 is at _____ place.

A. hundredth
B. ones
C. tenths
D. thousandths

36. In a tournament 8 teams were playing. Each team has to play one match with every other team. The total number of matches that were played are _____.

A. 25
B. 28
C. 56
D. 64

37. A dice is rolled twice. The probability of getting a sum of 6 on dice is _____

A. 1/36
B. 1/18
C. 5/36
D. 1/6

38. What is the angle between the hands of clock at 20 minutes to 8 O'clock?

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The sum of the series 2 + 2/3 + 2/9 + ... is _____.

A. 2
B. 3
C. 3.4
D. 3.5

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97. The number p is 4 more than 3 times the number r. The sum of number p and r is 10. Which of the following pair of equations could be used to find the values of p and r?

A. $p = 3r + 4; p + r = 10$
B. $p = 3r + 4; pr = 10$
C. $p = 3(r + 4); p + r = 10$
D. $p + 4 = 3r; p + r = 10$
E. $p + 4 = 3r; pr = 10$

98. What is the angle between two hands of the clock at 12'O clock?

A. 90°
B. 180°
C. 60°
D. 120°
E. 0°

99. Which of the following is the factor of $3x^3 - 11x^2 - 42x$?

A. $x - 7$
B. x
C. $x + 6$
D. $3x - 6$
E. $3x^2 - 11x$

100. A circle with a radius 3 feet has a circumference how many feet long?

A. 3π
B. 6π
C. 9π
D. 12π
E. $3/2\pi$

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Let $w = z^3 = 1$

$z^3 = 1$

$z^3 - 1 = 0 \Rightarrow z^3 - (1)^3 = 0$

$(z-1)(z^2+1+z) = 0$

$z = 1$

??? ??

$z^2 + z + 1 = 0$

$a = 1, b = 1, c = 1$

$x = \frac{-1 \pm \sqrt{1-4}}{2(1)}$

$= \frac{-1 \pm \sqrt{-3}}{2}$

$x = \frac{-1 \pm i\sqrt{3}}{2}$

1st Root $= w = \frac{-1 + \sqrt{3}i}{2}$

2nd Root $= w^2 = \frac{-1 - \sqrt{3}i}{2}$

option B

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34. The roots of the equation $x^2 + x + 1 = 0$ are _____.
(Where w is a cube root of 1).

A. 1, w, w^2
B. w, w^2
C. 1, w^2
D. 1, w

$x^2 + 1x - x + 1$

$x(x+1) \quad -1(x+1)$

$(x-1)(x+1)$

8

16

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